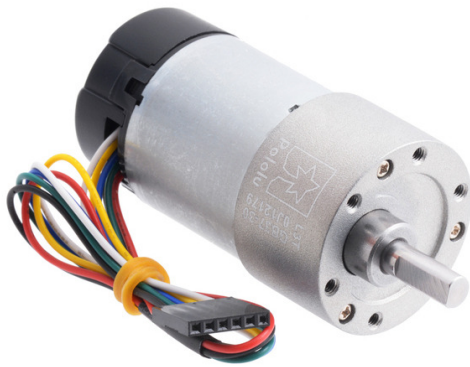


30:1 Metal Gearmotor 37Dx68L mm 12V with 64 CPR Encoder (Helical Pinion)



Pololu item #: 4752

Brand: [Pololu](#)

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Status: Active and Preferred

✓ RoHS 3

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Price break	Unit price (US\$)
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5	65.27
25	60.05
100	55.25

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This gearmotor is a powerful **12V** brushed DC motor with a **30:1** metal gearbox and an integrated quadrature encoder with a resolution of 64 counts per revolution (CPR) of the motor shaft and **1920 CPR** of the gearbox's output shaft. The gearbox is composed mainly of spur gears, but it features helical gears for the first stage for reduced noise and improved efficiency. These units have a 16 mm-long, 6 mm-diameter D-shaped output shaft. This gearmotor is also available [without an encoder](#).

Key specifications:

voltage	no-load performance	stall extrapolation
12 V	330 RPM, 200 mA	14 kg·cm (190 oz·in), 5.5 A

Alternatives available with variations in these parameter(s): gear ratio motor type encoders? [Select variant...](#)

[Description](#) [Specs \(22\)](#) [Pictures \(16\)](#) [Resources \(4\)](#) [FAQs \(3\)](#) [On the blog \(0\)](#) [Distributors \(63\)](#)

Overview

Measuring 37 mm (1.46") in diameter, these brushed DC gearmotors are the largest and most powerful we carry. They are available in a range of gear ratios from 6.3:1 to 150:1 and with 12 V or 24 V motors, and all versions are available with integrated 64 CPR quadrature encoders on the motor shafts. The 12 V and 24 V motors offer approximately the same performance at their respective nominal voltages, with the 24 V motor drawing half the current of the 12 V motor. See the [37D metal gearmotor datasheet](#) (2MB pdf) for more information, including detailed performance graphs for each gearmotor version. You can also use our dynamically sortable [37D gearmotor comparison table](#) to search for the version that offers the best combination of speed, torque, and current draw for your particular application. A more basic comparison table is available below.

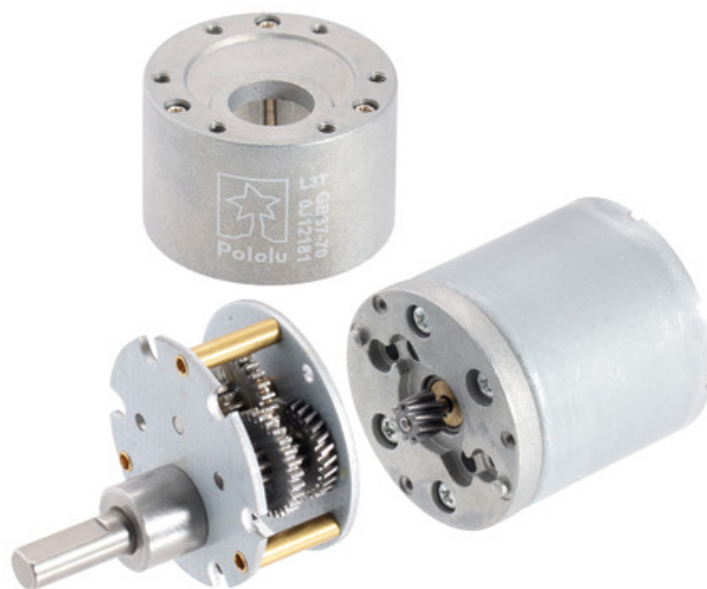
Rated Voltage	Stall Current	No-Load Current	Gear Ratio	No-Load Speed (RPM)	Extrapolated Stall Torque		Max Power (W)	 Without Encoder	 With Encoder
					(kg · cm)	(oz · in)			
12 V	5.5 A	0.2 A	1:1 (no gearbox)	10,000	0.5	7	–	–	item #4750
			6.3:1	1600	3.0	42	12	item #4747	item #4757
			10:1	1000	4.9	68	12	item #4748	item #4758
			19:1	530	8.5	120	12	item #4741	item #4751
			30:1	330	14	190	12	item #4742	item #4752
			50:1	200	21	290	10	item #4743	item #4753
			70:1	150	27	380	10*	item #4744	item #4754
			100:1	100	34	470	8*	item #4745	item #4755
			131:1	76	45	630	6*	item #4746	item #4756
			150:1	67	49	680	6*	item #2829	item #2828
24 V	3 A	0.1 A	1:1 (no gearbox)	10,000	0.55	8	–	–	item #4690
			6.3:1	1600	3.5	49	14	item #4688	item #4698
			10:1	1000	5.5	76	14	item #4689	item #4699
			19:1	530	9.5	130	13	item #4681	item #4691
			30:1	330	15	210	13	item #4682	item #4692
			50:1	200	23	320	12	item #4683	item #4693
			70:1	140	31	430	10*	item #4684	item #4694
			100:1	100	39	540	8*	item #4685	item #4695
			131:1	79	47	650	6*	item #4686	item #4696
			150:1	68	56	780	6*	item #4687	item #4697

* Output power for these units is constrained by gearbox load limits; spec provided is output power at max recommended load of 10 kg·cm.

Note: The listed stall torques and currents are theoretical extrapolations; units will typically stall well before these points as the motors heat up. Stalling or overloading gearmotors can greatly decrease their lifetimes and even result in immediate damage. The recommended upper limit for continuously applied loads is 10 kg·cm (150 oz·in), and the recommended upper limit for instantaneous torque is 25 kg·cm (350 oz·in). Stalls can also result in rapid (potentially on the order of seconds) thermal damage to the motor windings and brushes; a general recommendation for brushed DC motor operation is 25% or less of the stall current.

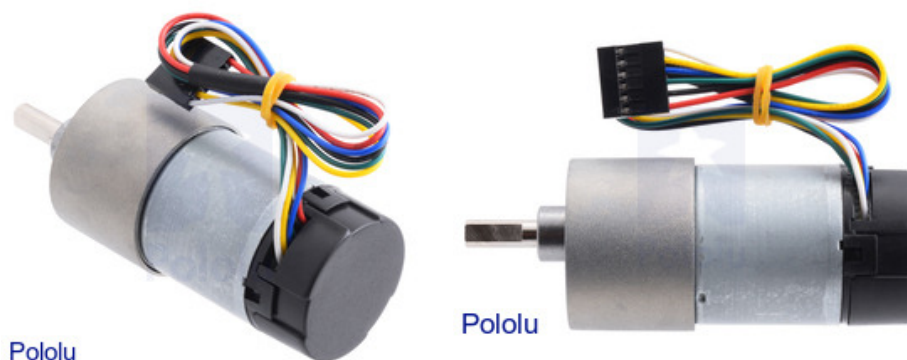
In general, these kinds of motors can run at voltages above and below the nominal voltages; lower voltages might not be practical, and higher voltages could start negatively affecting the life of the motor.

The gearboxes are composed mainly of spur gears, but they feature helical gears for the first stage for reduced noise and improved efficiency:

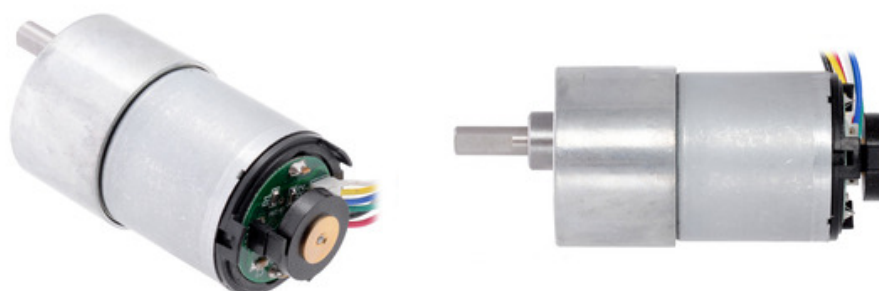


37D Gearmotor (Helical Pinion) with the gearbox removed showing the helical pinion gear and first mating gear.

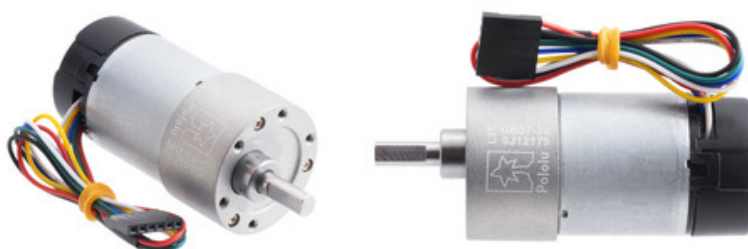
On units with encoders, the encoder portion is protected by a black plastic end cap:



The end cap is easily removable if you need to access the encoder or want to slightly reduce the overall gearmotor size, but there is a little bit of base plastic that will remain, as shown in the pictures below:



Details for item #4752





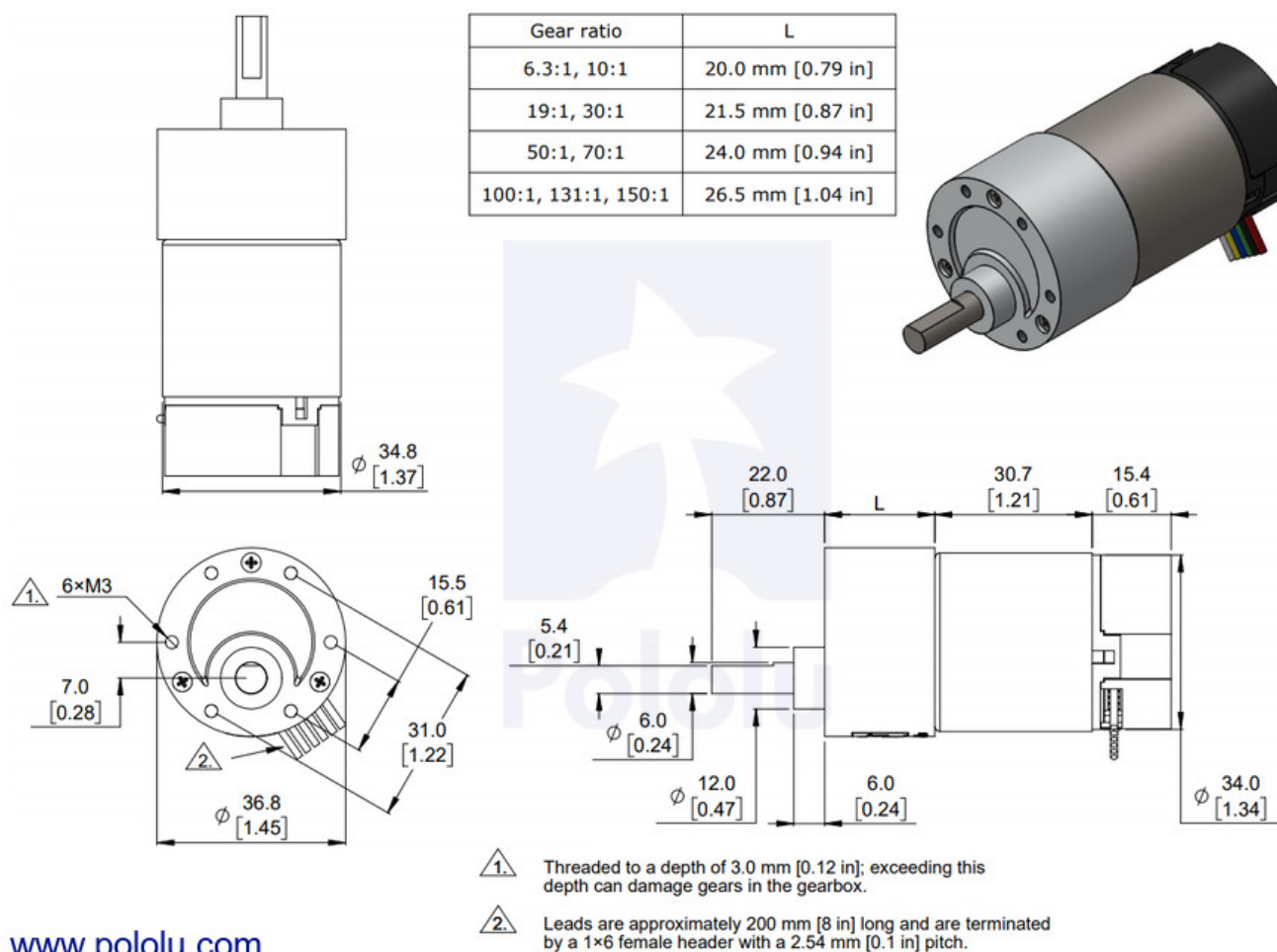
This gearmotor is a powerful **12V** brushed DC motor with a **30:1** metal gearbox and an integrated quadrature encoder that provides a resolution of 64 counts per revolution of the motor shaft, which corresponds to **1920 counts per revolution** of the gearbox's output shaft. These units have a 16 mm-long, 6 mm-diameter D-shaped output shaft. This gearmotor is also available [without an encoder](#).

Key specifications:

voltage	no-load performance	stall extrapolation
12 V	330 RPM, 200 mA	14 kg·cm (190 oz·in), 5.5 A

Exact gear ratio: $((25 \times 30 \times 40) / (10 \times 10 \times 10)) = 30:1$

Dimensions



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Dimensions of the 37D metal gearmotors with 64 CPR encoders. Units are mm over [inches].

This diagram is also available as a [downloadable PDF](#) (459k pdf).

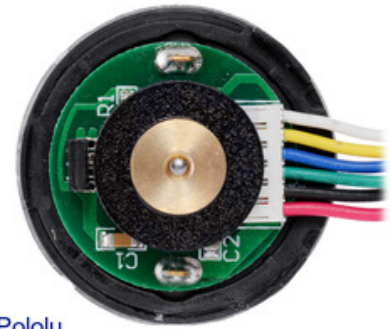
Warning: Do not screw too far into the mounting holes as the screws can hit the gears. We recommend screwing no more than 3mm (0.12") into the screw hole.

Using the Encoder

A two-channel Hall effect encoder is used to sense the rotation of a magnetic disk on a rear protrusion of the motor shaft. The quadrature encoder provides a resolution of 64 counts per revolution of the motor shaft when counting both edges of both channels. To compute the counts per revolution of the gearbox output, multiply the gear ratio by 64. The motor/encoder has six color-coded, 8" (20 cm) leads terminated by a 1×6 female header with a 0.1" pitch, as shown in the main product picture. This header works with standard [0.1" male headers](#) and our male [jumper](#) and [precrimped wires](#). If this header is not convenient for your application, you can pull the crimped wires out of the header or cut the header off. The following table describes the wire functions:

Color	Function
Red	motor power (connects to one motor terminal)
Black	motor power (connects to the other motor terminal)
Green	encoder GND
Blue	encoder Vcc (3.5 – 20 V)
Yellow	encoder A output
White	encoder B output

The Hall sensor requires an input voltage, Vcc, between 3.5 and 20 V and draws a maximum of 10 mA. The A and B outputs are square waves from 0 V to Vcc approximately 90° out of phase. The frequency of the transitions tells you the speed of the motor, and the order of the transitions tells you the direction. The following oscilloscope capture shows the A and B (yellow and white) encoder outputs using a 12 V motor at 12 V and a Hall sensor Vcc of 5 V:



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37D mm metal gearmotor with 64 CPR encoder (with end cap removed).



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Encoder A and B outputs for 37D mm metal gearmotor with 64 CPR encoder (12V motor running at 12 V).

By counting both the rising and falling edges of both the A and B outputs, it is possible to get 64 counts per revolution of the motor shaft. Using just a single edge of one channel results in 16 counts per revolution of the motor shaft, so the frequency of the A output in the above oscilloscope capture is 16 times the motor rotation frequency.

Gearmotor accessories

- **Mounting brackets:** The face plate has six mounting holes evenly spaced around the outer edge threaded for M3 screws. These mounting holes form a regular hexagon and the centers of neighboring holes are 15.5 mm apart. We carry two brackets for these gearmotors: a [stamped aluminum L-bracket](#) (sold in pairs) and a sturdier, tombstone-style [machined aluminum bracket](#) (sold individually):



37D mm gearmotor (without encoder) with L-bracket and 6mm universal mounting hub.



Pololu machined aluminum bracket for 37D mm metal gearmotors mounting a motor to a clear piece of acrylic.

- **Wheels and hubs:** The 6 mm diameter gearbox output shaft works with the [Pololu universal aluminum mounting hub for 6mm shafts](#), which can be used to mount our larger [Pololu wheels](#) (80mm- and

90mm-diameter) or custom wheels and mechanisms to the gearmotor's output shaft as shown in the left picture below. Alternatively, you could use our [6mm scooter wheel adapter](#) to mount many common scooter, skateboard, and inline skate wheels to the gearmotor's output shaft as shown in the right picture below:



Black Pololu Wheel 90×10mm on a Pololu 37D mm Metal Gearmotor.



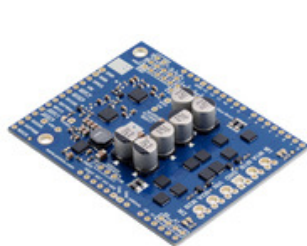
A 37D mm gearmotor connected to a scooter wheel by the 6 mm scooter wheel adapter.

For a general-purpose hex adapter, consider our [12mm hex wheel adapter](#) (also available in an [extended version](#)), which lets you use these motors with many common hobby RC wheels.

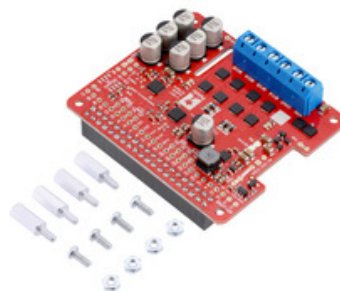


12mm Hex Wheel Adapter for 6mm Shaft connecting a Wild Thumper Wheel to a 37D mm Metal Gearmotor.

- **Motor controllers and drivers:** We have a number of [motor drivers](#) and [motor controllers](#) that work with these 37D mm metal gearmotors. We generally recommend our [High-Power Motor Drivers](#), which are available in various power levels and versions, including some [dual-channel shields for Arduino](#) and [dual-channel expansion boards for Raspberry Pi](#). Additionally, our VNH5019-based motor drivers, which are also available as [single](#) and [dual carriers](#), are a good match for these gearmotors.



Pololu Dual G2 High-Power Motor Driver 24v14 Shield for Arduino.

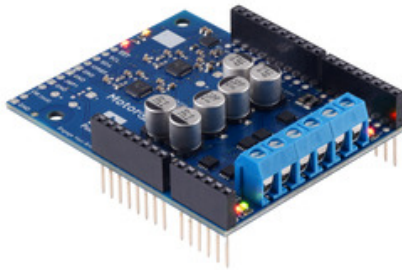


Pololu Dual G2 High-Power Motor Driver 24v14 for Raspberry Pi (assembled version) with included hardware.



Pololu dual VNH5019 motor driver shield for Arduino.

If you are looking for higher-level control interfaces, such as USB, RC, analog voltages, I²C, or TTL serial, consider our [Motoron motor controllers](#), [Jrk motor controllers](#), or [RoboClaw motor controllers](#); these controllers are available in various power levels several of which can handle the 37D mm metal gearmotors (we generally recommend a motor controller that can handle continuous currents above the stall current of your motor).



Motoron M2S18v18 Dual High-Power Motor Controller Shield for Arduino (Connectors Soldered).

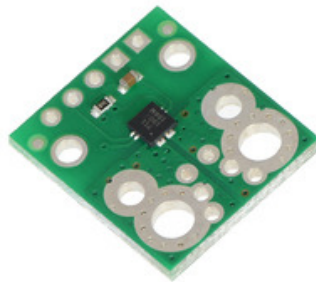


Jrk G2 21v3 USB Motor Controller with Feedback.



RoboClaw 2x7A Motor Controller (V5B) in its included case.

- **Current sensors:** We have an assortment of [current sensors](#) to choose from for those who need to monitor motor current:



ACS711EX current sensor carrier -15.5A to +15.5A.



ACS724LLCTR-05AB Current Sensor Carrier -5A to +5A.

Selecting the Right Gearmotor

We offer a wide selection of metal gearmotors that offer different combinations of speed and torque. Our [metal gearmotor comparison table](#) can help you find the motor that best meets your project's requirements.



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